

## CBSE Class 12 Chemistry — Half-Yearly Predicted Paper (Set 2 of 10)

### Section A: 1-Mark Questions (MCQs / Assertion-Reason)

- Q1.** The value of the van't Hoff factor ( $i$ ) for a dilute aqueous solution of  $K_2SO_4$ , assuming complete dissociation, is: (a) 1 (b) 2 (c) 3 (d) 4 [Solutions] [Confidence: Very High]
- Q2.** Which of the following halogens is typically exchanged in the Finkelstein reaction to form alkyl iodides? (a) Fluorine (b) Chlorine or Bromine (c) Only Bromine (d) Astatine [Haloalkanes and Haloarenes] [Confidence: Very High]
- Q3.** The unit of the rate constant for a second-order reaction is: (a)  $s^{-1}$  (b)  $mol\ L^{-1}\ s^{-1}$  (c)  $L\ mol^{-1}\ s^{-1}$  (d)  $L^2\ mol^{-2}\ s^{-1}$  [Chemical Kinetics] [Confidence: Very High]
- Q4.** Which of the following ions is diamagnetic in an aqueous solution? (a)  $Cr^{3+}$  (b)  $V^{3+}$  (c)  $Sc^{3+}$  (d)  $Fe^{2+}$  [The d- and f-Block Elements] [Confidence: Very High]
- Q5.** The IUPAC name of the complex  $K_3[Fe(CN)_6]$  is: (a) Potassium hexacyanidoferrate(II) (b) Potassium hexacyanidoferrate(III) (c) Potassium iron(III) hexacyanide (d) Tripotassium hexacyanoiron(III) [Coordination Compounds] [Confidence: Very High]
- Q6.** The reaction between an alkyl halide and sodium alkoxide to form an ether is called: (a) Wurtz reaction (b) Kolbe's reaction (c) Williamson's synthesis (d) Etard reaction [Alcohols, Phenols and Ethers] [Confidence: Very High]
- Q7.** Which of the following reagents is specifically used in the Rosenmund reduction of acid chlorides to aldehydes? (a)  $H_2/Pd - BaSO_4$  (b)  $Zn(Hg)/conc.\ HCl$  (c)  $NH_2NH_2/KOH$  (d)  $LiAlH_4$  [Aldehydes, Ketones and Carboxylic Acids] [Confidence: Very High]
- Q8.** During the electrolysis of aqueous  $NaCl$ , the gas liberated at the anode is: (a)  $O_2$  (b)  $H_2$  (c)  $Cl_2$  (d)  $HCl$  [Electrochemistry] [Confidence: High]
- Q9.** The reaction of a ketone with hydrazine ( $NH_2NH_2$ ) gives which of the following derivatives? (a) Oxime (b) Hydrazone (c) Semicarbazone (d) Cyanohydrin [Aldehydes, Ketones and Carboxylic Acids] [Confidence: High]
- Q10.** Which of the following is an example of an allylic halide? (a)  $CH_2 = CH - Cl$  (b)  $C_6H_5 - CH_2 - Cl$  (c)  $CH_2 = CH - CH_2 - Cl$  (d)  $C_6H_5 - Cl$  [Haloalkanes and Haloarenes] [Confidence: Very High]
- Q11.** What is the SI unit of cell constant ( $G^*$ )? (a)  $S\ cm^{-1}$  (b)  $cm^{-1}$  or  $m^{-1}$  (c)  $S\ cm^2\ mol^{-1}$  (d)  $\Omega\ cm$  [Electrochemistry] [Confidence: Very High]
- Q12.** Two solutions having the same osmotic pressure at a given temperature are called: (a) Isotonic solutions (b) Hypertonic solutions (c) Hypotonic solutions (d) Azeotropic solutions [Solutions] [Confidence: Very High]
- Directions for Q13 to Q16:** A statement of Assertion (A) followed by a statement of Reason (R) is given. Choose the correct answer out of the following choices: (a) Both A and R are true and R is the correct explanation of A. (b) Both A and R are true but R is not the correct explanation of A. (c) A is true but R is false. (d) A is false but R is true.
- Q13.** Assertion (A): Elevation in boiling point is a colligative property. Reason (R): It depends only on the number of solute particles in the solution, regardless of their nature. [Solutions] [Confidence: Very High]
- Q14.** Assertion (A): The rate of a zero-order reaction is independent of the initial concentration of the reactants. Reason (R): The half-life of a zero-order reaction is independent of the initial concentration of the reactants. [Chemical Kinetics] [Confidence: High]
- Q15.** Assertion (A): Ethers generally have lower boiling points than isomeric alcohols. Reason (R): Ethers do not have a hydrogen atom directly bonded to oxygen and cannot form intermolecular hydrogen bonds. [Alcohols, Phenols and Ethers] [Confidence: Very High]
- Q16.** Assertion (A): Zirconium ( $Zr$ ,  $Z = 40$ ) and Hafnium ( $Hf$ ,  $Z = 72$ ) possess almost similar atomic radii. Reason (R): This is due to the effect of lanthanoid contraction. [The d- and f-Block Elements] [Confidence: Very High]

### Section B: 2-Mark Questions (Short Answer)

- Q17.** State Henry's law and write its mathematical expression. Mention any one important biological or industrial application of Henry's law. [Solutions] [Confidence: Very High]

- Q18.** Using IUPAC norms, write the exact formulae for the following: (i) Sodium dicyanoaurate(I) (ii) Hexaammineplatinum(IV) chloride [Coordination Compounds] [Confidence: High]
- Q19.** Define a pseudo-first-order reaction. Give one chemical example to illustrate it. [Chemical Kinetics] [Confidence: Very High]
- Q20.** Write the mechanism for the acid-catalysed dehydration of ethanol to yield ethoxyethane (diethyl ether) at 413 K. [Alcohols, Phenols and Ethers] [Confidence: Very High]
- Q21.** Why do transition metals frequently form interstitial compounds? State one physical property of these compounds that differs from the parent metal. [The d- and f-Block Elements] [Confidence: High]

### Section C: 3-Mark Questions (Short Answer)

- Q22.** Compound A ( $C_4H_9Br$ ) is treated with aqueous  $KOH$  solution. The rate of this reaction depends only upon the concentration of A. When another isomeric compound B of the same formula is treated with aqueous  $KOH$ , the rate of reaction depends on the concentration of both the compound and  $KOH$ . (i) Identify the structure and IUPAC name of Compound A. (ii) Identify the structure and IUPAC name of Compound B. (iii) Name the specific reaction mechanism followed by B. [Haloalkanes and Haloarenes] [Confidence: Very High]
- Q23.** The resistance of a conductivity cell filled with  $0.1 \text{ mol L}^{-1} KCl$  solution is  $100 \Omega$ . If the resistance of the same cell when filled with  $0.02 \text{ mol L}^{-1} KCl$  solution is  $520 \Omega$ , calculate the conductivity and molar conductivity of the  $0.02 \text{ mol L}^{-1} KCl$  solution. (Given: The conductivity of  $0.1 \text{ mol L}^{-1} KCl$  solution is  $1.29 \times 10^{-2} S \text{ cm}^{-1}$ ). [Electrochemistry] [Confidence: Very High]
- Q24.** How will you seamlessly convert the following in not more than two steps? (i) Propanone to Propene (ii) Benzene to Acetophenone (iii) Ethanoic acid to 2-Chloroethanoic acid [Aldehydes, Ketones and Carboxylic Acids] [Confidence: High]
- Q25.** State any three points of difference between a non-ideal solution showing positive deviation and a non-ideal solution showing negative deviation from Raoult's Law. Provide one example of each. [Solutions] [Confidence: High]
- Q26.** Complete and balance the following chemical equations: (i)  $Cr_2O_7^{2-} + Sn^{2+} + H^+ \rightarrow$  (ii)  $MnO_4^- + I^- + H_2O \rightarrow$  (in neutral/faintly alkaline medium) (iii)  $2MnO_2 + 4KOH + O_2 \xrightarrow{\Delta}$  [The d- and f-Block Elements] [Confidence: Very High]
- Q27.** Draw the structures of the geometrical isomers of the complex  $[Pt(en)_2Cl_2]^{2+}$ . Which of these isomers is optically active and why? [Coordination Compounds] [Confidence: Very High]
- Q28.** A first-order reaction is 40% completed in 80 minutes. Calculate the value of its rate constant ( $k$ ). (Given:  $\log 10 = 1$ ,  $\log 2 = 0.301$ ,  $\log 3 = 0.477$ ). [Chemical Kinetics] [Confidence: High]

### Section D: 4-Mark Questions (Case-Study)

- Q29. Case Study: Colligative Properties and Freezing Point Depression** Boiling point or freezing point of a liquid solution is heavily affected by dissolved non-volatile solids in the liquid phase. A soluble solid in solution has the effect of raising its boiling point and depressing its freezing point. The decrease in freezing point is directly correlated to the concentration of solutes dissolved in the solvent. This phenomenon, known as freezing point depression, is a colligative property useful for applications such as freeze-concentration of liquid food and melting ice on roads. (a) What is the value of the Van't Hoff factor ( $i$ ) for  $K_3[Fe(CN)_6]$ , assuming complete dissociation in water? (1 Mark) (b) Why is Calcium Chloride ( $CaCl_2$ ) widely used to clear snow from roads in cold countries? (1 Mark) (c) Calculate the freezing point of an aqueous solution containing 10.5 g of Magnesium Bromide ( $MgBr_2$ ) in 200 g of water, assuming complete dissociation. (Given: Molar mass of  $MgBr_2 = 184 \text{ g mol}^{-1}$ ,  $K_f$  for water =  $1.86 \text{ K kg mol}^{-1}$ , freezing point of pure water =  $0^\circ C$ ). (2 Marks) [Solutions] [Confidence: Very High]
- Q30. Case Study: Nucleophilic Substitution in Haloalkanes** Alkyl halides undergo nucleophilic substitution reactions rapidly. The mechanism can follow either an  $S_N1$  pathway (unimolecular) or an  $S_N2$  pathway (bimolecular) depending strictly on the structure of the alkyl halide, the polarity of the solvent, and the strength of the incoming nucleophile.  $S_N1$  reactions proceed via a carbocation intermediate and lead to racemization if the alkyl halide is chiral.  $S_N2$  reactions involve a concerted transition state and lead to complete inversion of stereochemistry. (a) Out of

chloromethane ( $\text{CH}_3\text{Cl}$ ) and fluoromethane ( $\text{CH}_3\text{F}$ ), which one possesses a higher dipole moment? (1 Mark) (b) Explain chemically why the  $\text{S}_{\text{N}}1$  reaction of an optically active alkyl halide strictly leads to racemization. (1 Mark) (c) Predict the major product of the reaction between tert-butyl bromide and concentrated sodium methoxide ( $\text{CH}_3\text{ONa}$ ). Name the exact type of reaction mechanism taking place here instead of substitution. (2 Marks) [Haloalkanes and Haloarenes] [Confidence: High]

### Section E: 5-Mark Questions (Long Answer)

**Q31.** (a) State Kohlrausch's law of independent migration of ions. (1 Mark) (b) Calculate the limiting molar conductivity ( $\Lambda_m^\circ$ ) for Magnesium Chloride ( $\text{MgCl}_2$ ) if the limiting ionic conductivities are  $\lambda^\circ(\text{Mg}^{2+}) = 106.0 \text{ S cm}^2 \text{ mol}^{-1}$  and  $\lambda^\circ(\text{Cl}^-) = 76.3 \text{ S cm}^2 \text{ mol}^{-1}$ . (2 Marks) (c) A steady current of  $2.0 \text{ A}$  was passed through a solution of Silver Nitrate ( $\text{AgNO}_3$ ) for exactly **15 minutes**. Calculate the mass of silver (Ag) deposited at the cathode. (Given: Molar mass of  $\text{Ag} = 108 \text{ g mol}^{-1}$ ,  $1\text{F} = 96500 \text{ C mol}^{-1}$ ). (2 Marks) [Electrochemistry] [Confidence: Very High]

**Q32.** (a) Give reliable, simple chemical tests to uniquely distinguish between the following pairs of compounds: (i) Pentan-2-one and Pentan-3-one (ii) Ethanal and Propanal (2 Marks) (b) Describe the following important named organic reactions with a suitable balanced chemical equation for each: (i) Wolff-Kishner reduction (ii) Clemmensen reduction (iii) Cross Aldol Condensation (using suitable aliphatic examples). (3 Marks) [Aldehydes, Ketones and Carboxylic Acids] [Confidence: Very High]

**Q33.** (a) Write the detailed mechanism for the reaction of Hydrogen Iodide ( $\text{HI}$ ) with methoxymethane ( $\text{CH}_3 - \text{O} - \text{CH}_3$ ) to form methyl iodide and methanol. (2 Marks) (b) Explain precisely why phenol is considerably more acidic than ethanol. (2 Marks) (c) Write the balanced chemical equation for the Reimer-Tiemann reaction. (1 Mark) [Alcohols, Phenols and Ethers] [Confidence: Very High]

### Answer Key

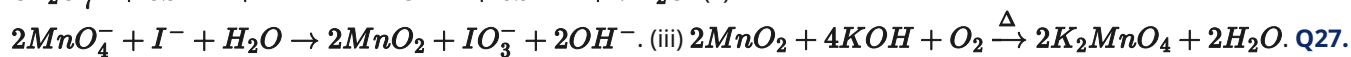
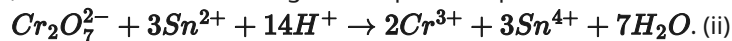
**Q1.** (c) 3. ( $\text{K}_2\text{SO}_4 \rightarrow 2\text{K}^+ + \text{SO}_4^{2-}$ ). **Q2.** (b) Chlorine or Bromine. (R-Cl/R-Br reacts with NaI to give R-I). **Q3.** (c)  $\text{L mol}^{-1} \text{ s}^{-1}$ . **Q4.** (c)  $\text{Sc}^{3+}$ . (It has a  $3d^0$  configuration with no unpaired electrons). **Q5.** (b) Potassium hexacyanidoferrate(III). **Q6.** (c) Williamson's synthesis. **Q7.** (a)  $\text{H}_2/\text{Pd} - \text{BaSO}_4$ . **Q8.** (c)  $\text{Cl}_2$ . (Due to the overpotential of oxygen, chloride ions are preferentially oxidized). **Q9.** (b) Hydrazone. **Q10.** (c)  $\text{CH}_2 = \text{CH} - \text{CH}_2 - \text{Cl}$ . **Q11.** (b)  $\text{cm}^{-1}$  or  $\text{m}^{-1}$ . **Q12.** (a) Isotonic solutions. **Q13.** (a) Both A and R are true and R is the correct explanation of A. **Q14.** (c) A is true but R is false. (For a zero-order reaction,  $t_{1/2} = [\text{R}]_0/2k$ , which is dependent on initial concentration). **Q15.** (a) Both A and R are true and R is the correct explanation of A. **Q16.** (a) Both A and R are true and R is the correct explanation of A.

**Q17.** Henry's Law states that the partial pressure of a gas in the vapor phase ( $p$ ) is proportional to the mole fraction of the gas ( $x$ ) in the solution.  $p = K_H \cdot x$ . Application: Used in SCUBA diving tanks (diluting air with He) to avoid "the bends" or in the bottling of carbonated beverages under high pressure to increase  $\text{CO}_2$  solubility. **Q18.** (i)  $\text{Na}[\text{Au}(\text{CN})_2]$  (ii)  $[\text{Pt}(\text{NH}_3)_6]\text{Cl}_4$ . **Q19.** Reactions which are not truly of the first order but behave as first-order reactions under certain conditions (e.g., when one reactant is in large excess) are pseudo-first-order reactions.

Example: Acid-catalyzed hydrolysis of ethyl acetate:  $\text{CH}_3\text{COOC}_2\text{H}_5 + \text{H}_2\text{O} \xrightarrow{\text{H}^+} \text{CH}_3\text{COOH} + \text{C}_2\text{H}_5\text{OH}$ . **Q20.** Step 1: Protonation of alcohol:  $\text{CH}_3\text{CH}_2\text{OH} + \text{H}^+ \rightleftharpoons \text{CH}_3\text{CH}_2\text{OH}_2^+$ . Step 2: Nucleophilic attack by unprotonated alcohol ( $\text{S}_{\text{N}}2$ ):  $\text{CH}_3\text{CH}_2\text{OH} + \text{CH}_3\text{CH}_2\text{OH}_2^+ \rightarrow \text{CH}_3\text{CH}_2 - \text{O}^+(\text{H}) - \text{CH}_2\text{CH}_3 + \text{H}_2\text{O}$ . Step 3: Deprotonation:  $\text{CH}_3\text{CH}_2 - \text{O}^+(\text{H}) - \text{CH}_2\text{CH}_3 \rightarrow \text{CH}_3\text{CH}_2 - \text{O} - \text{CH}_2\text{CH}_3 + \text{H}^+$ . **Q21.** Transition metals form interstitial compounds because small non-metal atoms (like H, C, or N) easily fit into the empty spaces (interstices) of their crystal lattices. Physical property: They are much harder than the pure metals and have higher melting points.

**Q22.** (i) A undergoes  $\text{S}_{\text{N}}1$  (rate depends only on alkyl halide), so A is a tertiary halide: 2-Bromo-2-methylpropane (tert-butyl bromide). (ii) B undergoes  $\text{S}_{\text{N}}2$  (rate depends on halide and nucleophile), so B is a primary halide: 1-Bromobutane. (iii)  $\text{S}_{\text{N}}2$  (Bimolecular Nucleophilic Substitution). **Q23.** Cell constant  $G^* = \kappa \times R = 1.29 \times 10^{-2} \text{ S cm}^{-1} \times 100 \Omega = 1.29 \text{ cm}^{-1}$ . Conductivity of  $0.02\text{M}$  ( $\kappa_2$ ) =  $G^*/R_2 = 1.29/520 = 2.48 \times 10^{-3} \text{ S cm}^{-1}$ . Molar conductivity

$\Lambda_m = (\kappa \times 1000)/C = (2.48 \times 10^{-3} \times 1000)/0.02 = 124 \text{ S cm}^2 \text{ mol}^{-1}$ . **Q24.** (i) Propanone  $\xrightarrow{\text{LiAlH}_4 \text{ or } \text{NaBH}_4}$  Propan-2-ol  $\xrightarrow{\text{Conc. H}_2\text{SO}_4, 443\text{K}}$  Propene. (ii) Benzene +  $\text{CH}_3\text{COCl} \xrightarrow{\text{Anhy. AlCl}_3}$  Acetophenone +  $\text{HCl}$  (Friedel-Crafts Acylation). (iii)  $\text{CH}_3\text{COOH} + \text{Cl}_2 \xrightarrow{\text{Red P}} \text{Cl}-\text{CH}_2-\text{COOH}$  (Hell-Volhard-Zelinsky reaction). **Q25.** Positive Deviation: A-B interactions are weaker than A-A/B-B;  $\Delta H_{\text{mix}} > 0$ ;  $\Delta V_{\text{mix}} > 0$ ; Forms minimum boiling azeotrope. Example: Ethanol + Acetone. Negative Deviation: A-B interactions are stronger than A-A/B-B;  $\Delta H_{\text{mix}} < 0$ ;  $\Delta V_{\text{mix}} < 0$ ; Forms maximum boiling azeotrope. Example: Chloroform + Acetone. **Q26.** (i)



Draw square planar projection for octahedral complex.  $\text{cis-}[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$  has two  $\text{Cl}^-$  adjacent ( $90^\circ$ ).  $\text{trans-}[\text{Pt}(\text{en})_2\text{Cl}_2]^{2+}$  has two  $\text{Cl}^-$  opposite ( $180^\circ$ ). The *cis*-isomer is optically active because it lacks a plane of symmetry, making its mirror image non-superimposable. (Trans has a plane of symmetry). **Q28.** For a first-order reaction:

$$k = \frac{2.303}{t} \log \left( \frac{[R]_0}{[R]} \right). \text{ 40\% complete means 60\% remains. } k = \frac{2.303}{80} \log \left( \frac{100}{60} \right) = \frac{2.303}{80} \log(1.667). \text{ Alternatively:}$$

$$\log(10/6) = \log 10 - \log(2 \times 3) = 1 - (0.301 + 0.477) = 1 - 0.778 = 0.222.$$

$$k = \frac{2.303}{80} \times 0.222 = 0.00639 \text{ min}^{-1}.$$

**Q29.** (a)  $i = 4$ . ( $\text{K}_3[\text{Fe}(\text{CN})_6] \rightarrow 3\text{K}^+ + [\text{Fe}(\text{CN})_6]^{3-}$ ). (b)  $\text{CaCl}_2$  significantly depresses the freezing point of water well below environmental temperatures, causing ice to melt into a liquid solution. (c) Moles of  $\text{MgBr}_2 = 10.5/184 = 0.057 \text{ mol}$ . Molality ( $m$ ) =  $0.057/0.200 \text{ kg} = 0.285 \text{ m}$ .  $i$  for  $\text{MgBr}_2 = 3$ .

$$\Delta T_f = i \times K_f \times m = 3 \times 1.86 \times 0.285 = 1.59^\circ\text{C}. \text{ Freezing point of solution } T_f = 0^\circ\text{C} - 1.59^\circ\text{C} = -1.59^\circ\text{C}.$$

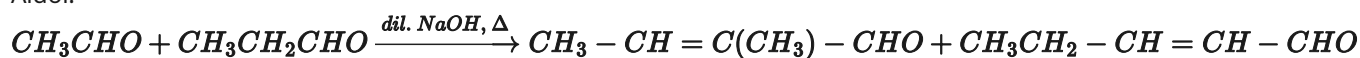
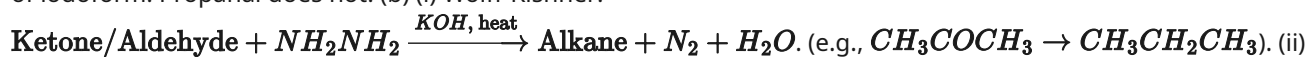
**Q30.** (a) Chloromethane ( $\text{CH}_3\text{Cl}$ ). The  $\text{C}-\text{Cl}$  bond length is larger than  $\text{C}-\text{F}$ , compensating for F's higher electronegativity ( $\mu = q \times d$ ). (b) In  $\text{S}_\text{N}1$ , the leaving group departs to form a flat, planar carbocation intermediate. The nucleophile can then attack this planar carbocation from either face (top or bottom) with equal 50/50 probability, creating an equimolar racemic mixture of enantiomers. (c) Major product: 2-Methylpropene (Isobutylene). Type of reaction:  $\beta$ -Elimination (Dehydrohalogenation). Bulky tert-butyl halides heavily favor elimination (E2) over substitution when treated with a strong base like methoxide.

**Q31.** (a) Kohlrausch's law states that the limiting molar conductivity of an electrolyte can be represented as the sum of the individual contributions of the anion and cation of the electrolyte. (b)

$$\Lambda_m^\circ(\text{MgCl}_2) = \lambda^\circ(\text{Mg}^{2+}) + 2 \times \lambda^\circ(\text{Cl}^-) = 106.0 + 2(76.3) = 106.0 + 152.6 = 258.6 \text{ S cm}^2 \text{ mol}^{-1}. \text{ (c)}$$

Charge  $Q = I \times t = 2.0 \text{ A} \times (15 \times 60) \text{ s} = 1800 \text{ C}$ . Reaction:  $\text{Ag}^+ + \text{e}^- \rightarrow \text{Ag}$ . ( $n = 1$ ). Mass =

$$\frac{M \times Q}{n \times F} = \frac{108 \times 1800}{1 \times 96500} = 2.014 \text{ g. Q32. (a) (i) Iodoform test: Pentan-2-one has a methyl ketone group and gives a yellow precipitate (CHI}_3) \text{ with NaOH/I}_2. \text{ Pentan-3-one does not. (ii) Iodoform test: Ethanal (CH}_3\text{CHO) gives a yellow ppt of iodoform. Propanal does not. (b) (i) Wolff-Kishner:}$$



(plus self-aldol products). **Q33.** (a) Step 1: Protonation of ether oxygen to form an oxonium ion:

